



The price responsiveness of housing supply in OECD countries



Aida Caldera^{*}, Åsa Johansson

OECD, Economics Department, 2 Rue André Pascal, 75775 Paris CEDEX 16, France

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ABSTRACT

The responsiveness of housing supply to changes in prices bears important implications for the evolution of housing prices and the speed of adjustment of housing markets. Based on a stock-flow model of the housing market estimated within an error correction framework, this paper estimates the long-run price elasticity of new housing supply in 21 OECD countries. Estimates suggest that the responsiveness of housing supply to price changes varies substantially across countries. It is relatively more flexible in North America and some Nordic countries, while it is more rigid in continental European countries and in the United Kingdom. The responsiveness of housing supply depends not only on national geographical and urban characteristics but also on policies, such as land use and planning regulations.

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1. Introduction

The responsiveness of housing supply to changes in prices is a crucial factor in the functioning of housing markets. It determines the extent to which the housing market responds to demand shocks with increased construction or higher prices. This has potential implications for the evolution of key housing market outcomes such as housing prices and affordability. For instance, existing evidence suggests that in supply-constrained markets, most of the adjustment occurs in the price of housing rather than in expanding housing supply (Glaeser et al., 2008; Gyourko, 2009). Other evidence suggests that regions with high supply responsiveness have relatively small price rises following demand shocks (e.g. Grimes and Aitken, 2006). Housing price “bubbles” are more common and last longer in areas where supply is inelastic Glaeser et al. (2008). Supply conditions also matter for house price volatility and aggregate economic

stability. An unresponsive housing supply can increase the sensitivity of house prices to demand shocks and, thus, influence private consumption patterns and residential investment. For instance, in the short to medium term, an increase in housing demand will translate into lower increases in real house prices in areas with more responsive housing supply. However, the flip side is that in flexible-supply areas, housing investment adjusts more rapidly to greater changes in demand contributing to more cyclical swings in economic growth, as witnessed by recent developments. A quantification of the responsiveness of housing supply with respect to prices can, therefore, shed light on the trend evolution and volatility of house prices in OECD countries. It can also provide valuable information for housing policy reforms aimed at dampening housing price volatility and increasing macroeconomic resilience to shocks.

Despite its importance for the analysis of housing markets and policy, very little cross-country empirical evidence exists on the responsiveness of supply with respect to prices, partly reflecting data constraints. This paper aims to fill this gap, by estimating the long-run price elasticity of new housing supply for 21 OECD countries for which data

^{*} Corresponding author. Fax: +33 1 44 30 63 83.

E-mail addresses: aida.calderasanchez@oecd.org (A. Caldera), Asa.Johansson@oecd.org (Å. Johansson).

are available. The analysis is based on a stock-flow model of the housing market. The price-elasticity of new housing supply is estimated separately for each country over the period from the early 1980s to mid-2000s, as the price coefficient on a long-run supply equation, which is jointly estimated with a long-run price equation in an error correction framework.¹ This empirical approach has the advantage of being simple enough, so that comparable estimates can be obtained for a large number of OECD countries, and be grounded on sound theoretical foundations. Ample evidence suggests that housing markets only adjust slowly to changes in market conditions due to, among other factors, product heterogeneity and costly search and transaction costs. The error correction framework employed in this paper takes into account such slow adjustment of housing markets.

The empirical results suggest that housing responsiveness varies substantially across countries, with potential consequences for the speed of adjustment of housing markets. New housing supply tends to be relatively flexible in North America and some Nordic countries, while it is more rigid in continental European countries and in the United Kingdom. For instance, long-run supply elasticities in the United States, Canada, Sweden and Denmark are above unity, implying that in response to a demand shock, housing output will increase relatively more than prices. On the other hand, the responsiveness of housing supply tends to be quite low in countries such as Switzerland, the Netherlands, Austria or Italy. These estimates are broadly in line with casual and early evidence (e.g. Swank et al., 2002). The responsiveness of housing supply depends not only on national geographical and urban characteristics but also on policies, such as land use, planning and rental regulations. While the link is hard to establish empirically due to data limitations, we present some suggestive evidence that cumbersome land use and planning regulations across OECD countries are associated with a less responsive housing supply in the long-run.

The remainder of the paper is organised as follows. Section 2 presents the theoretical framework used to analyse the housing sector. Section 3, describes the approach to estimate the price-elasticity of new housing supply and the data. Section 4 presents the empirical results and compares the estimates of the responsiveness of housing supply with those of other studies. Section 5 discusses potential drivers of differences in the responsiveness of housing supply across countries. Section 6 concludes and discusses policy options to enhance the responsiveness of housing supply.

2. A model of the housing sector

This section presents the conceptual framework underlying the estimation of the long-run price elasticity of new housing supply. The empirical approach builds on a stock-flow model of the housing sector to estimate the elasticity

of new housing supply with respect to price. This model has long been used in the literature as the main theoretical framework to describe the housing sector (see Di Pasquale and Wheaton (1994) for a review). It has the key advantage of considering the dual role of housing as a capital investment and consumption good, as well as to distinguish between the stock of housing and the flow of housing investment. One important feature of the housing market is that the housing stock adjusts slowly to changes in demand: housing investment is lumpy as building takes time and depreciation of the housing stock is slow. Thus, housing markets can clear rapidly only if prices react strongly to tensions between demand and supply. However, the heterogeneity of housing generates search and transaction costs which make it difficult for households to react swiftly to price signals (Di Pasquale and Wheaton, 1994). Hence, stock equilibrium is achieved only in the long-run.

In the long-run, two important drivers will influence housing demand (Meen, 2002). First, expected or permanent income and developments in the age and structure of the population will determine the quantity of housing demanded. Second, the user cost of holding the housing asset, which is in turn influenced by interest rates, taxation and expected capital gains of owning the house also matters for demand. Housing demand can be expressed as a function of exogenous factors such as, income, demographic characteristics and the user cost of housing (summarised in X_1), and the real price level of housing (P). In the long run, the demand for housing (D) determines the equilibrium price that will clear the stock of housing (S), as given by the following equation:

$$D(X_1, P) = S = \int_0^T dS \quad (1)$$

On the supply side, the stock of housing slowly depreciates over time, at a depreciation rate ∂ , and expands gradually with new residential investment (I) as given by the differential Eq. (2):

$$dS = I(X_2, P) - \partial S \quad (2)$$

The stock equilibrium demand, given by Eq. (1) leads to long-run adjustments in the rate of growth of the housing stock through investment in new construction. Residential investment is, in turn, a function of cost shifters, such as costs of production in terms of land, labour and materials, and housing policies summarised in the vector X_2 , as well as real house prices. Because house prices affect the incentives to build new housing and/or maintain the existing stock of housing they have an effect on residential investment.

Inverting the demand function (1) (assuming a log-linear relationship between the variables) and adding an error term ε_t^p , the observed price, p_t can be expressed as a function of the equilibrium price p_t^* (determined by the housing stock and demand shifters):

$$p_t = \alpha_0 - \alpha_1 s_t + \alpha_2' X_1 + \varepsilon_t^p = p_t^* + \varepsilon_t^p \quad (3)$$

where all the variables are expressed in logs (and denoted in small letters) and the subscript t denotes time. Similarly, long-run residential investment can be expressed as

¹ The analysis is macro in nature, essentially treating each country as a single housing market. Although in reality housing markets are typically local or regional, country-level estimates give a sense of differences in the overall responsiveness of housing supply across countries.

follows, where the coefficient β_1 denotes the long-run elasticity of investment with respect to prices²:

$$i_t = \beta_0 + \beta_1 p_t + \beta_2 X_2 + \varepsilon_t^i = i_t^* + \varepsilon_t^i \quad (4)$$

Taking into account the slow adjustment of housing markets the model further incorporates an error correction process in both the demand and supply sides. Ample evidence suggests that the housing market adjusts gradually to shocks due to, among other factors, transaction costs (e.g. Di Pasquale and Wheaton, 1994; Mayer and Somerville, 2000). For instance, Di Pasquale and Wheaton (1994) suggest that product heterogeneity and costly searches imply that the anticipated time for selling a house can be long and exhibit significant variance leading to a slow clearing of the housing market. Specifically, if the housing market is in equilibrium and a shock occurs, this creates a disequilibrium causing a wedge between the equilibrium price and the actual price level ($p_t - p_t^*$) due to such rigidities. For instance, if the desired level of the housing stock is higher than the existing supply, a negative wedge will appear between the actual housing and equilibrium price, and actual house prices will gradually rise spurring new residential investment until the equilibrium levels are reached.

The following short-run demand equation summarises the dynamics of house prices and their reaction to housing market imbalances:

$$\Delta p_t = \phi_0 + \phi_1 \varepsilon_{t-1}^p + \phi_2' \Delta X_1 + u_t^p \quad (5)$$

where Δ denotes the first difference of a variable and ε_t^p captures the disequilibrium between actual and equilibrium house price. The speed of price adjustment to equilibrium is captured by ϕ_1 and is expected to be negative reflecting the convergence between actual prices and equilibrium prices over time. Furthermore, changes in house prices are a function of changes in other factors such as household income or the user cost, summarised in X_1 and u_t^p is an error term.

The dynamics of residential investment are, in turn, described by the following equation, where the error correction term ε_t^i tracks the level of over or under investment.

$$\Delta i_t = \delta_0 + \delta_1 \varepsilon_{t-1}^i + \delta_2' \Delta X_2 + u_t^i \quad (6)$$

Eq. (6) indicates that a fraction δ_1 of the deviations in investment from equilibrium are corrected over the following period, at a speed of adjustment δ_1 (with $\delta_1 < 0$ to ensure convergence). The vector X_2 in turn captures factors which influence changes in residential investment, such as variations in the costs of construction and u_t^i is an error term.

3. Estimation and data

3.1. Econometric model

Building on the theoretical framework outlined above and on work by Hüfner and Lundsgaard (2007) and Rae

and van den Noord (2006), the following long-run price and investment equations are estimated in an error correction framework employing the Engle and Granger (1987) two-step estimation procedure. In order to obtain an estimate of the long-run price-elasticity of new housing supply for each country, the following system of equations is estimated separately for each OECD country using quarterly data:

$$P_t = \alpha_0 + \alpha_1 y_t + \alpha_2 R_t + \alpha_3 s_t + \alpha_4 d_t + \gamma_t + ECT_t^p \quad (7)$$

$$i_t = \beta_0 + \beta_1 p_{t-1} + \beta_2 cc_{t-1} + \beta_3 d_t + \gamma_t + ECT_t^i \quad (8)$$

The dependent variable in the demand equation is real house prices, p_t , while the dependent variable in the supply equation is real gross residential investment, i_t . With respect to prices, the coefficient of interest in these equations is β_1 , which measures the long-run elasticity of housing investment, or new housing supply. As a robustness check the investment equations were also estimated separately from the price equation. The resulting elasticity estimates and the country ranking in terms of responsiveness are in line with system results. The system approach is preferred as it allows for feedback between supply and demand and is a more efficient estimation approach.

The explanatory variables in the price equation include real income (y_t), the real interest rate (R_t), the stock of residential dwellings (s_t), and a demographic variable (d_t) capturing the share of the age cohort 25–44 in population, i.e. those who are more likely to buy housing. The real interest rate is a simple measure of the user cost, which measures the opportunity cost of capital tied up in the property or taken on credit. The coefficients on the stock of housing and interest rates are expected to be negative, while the coefficient on income and demographics are expected to be positive. The explanatory variables in the investment equation include real residential construction costs (cc_{t-1}), real house prices (p_{t-1}) and the same demographic variables included in the first equation (d_t), as the size and structure of the population is expected to influence the incentives to build. The coefficients on construction costs are expected to be negative, while the coefficients on prices and the demographic variable are expected to be positive. All the variables are in logs, except the real interest rate. Both construction costs and real house prices enter lagged in the equation to reflect the nature of the construction industry, where there is typically a lag between price signals and investment in housing, and to avoid potential endogeneity. In addition, both equations include a set of quarterly dummies γ_t to control for seasonal effects.

The estimated residuals ECT_t^p and ECT_t^i are then included as error correction terms, in the following equations explaining the short-term evolution of prices and investment:

$$\Delta p_t = \phi_0 + \phi_1 \Delta y_t + \phi_2 \Delta R_t + \phi_3 \Delta s_t + \phi_4 \Delta d_t + \phi_5 ECT_{t-1}^p + \gamma_t + \varepsilon_t \quad (9)$$

$$\Delta i_t = \delta_0 + \delta_1 \Delta cc_{t-1} + \delta_2 \Delta p_{t-1} + \delta_3 \Delta d_t + \delta_4 ECT_{t-1}^i + \gamma_t + v_t \quad (10)$$

² The β_1 parameter can be interpreted as the elasticity of the flow of residential investment (or new supply) with respect to prices. The elasticity β_1 divided by the depreciation rate (ϑ) of the housing stock.

where ECT_{t-1} is the error-correction term, i.e. the residual from the long-run equation, lagged one period. The coefficients ϕ_5 and δ_4 measure the (quarterly) speed of adjustment to the long term equilibrium. The parameter ϕ_5 is expected to be negative, as disequilibrium in prices in previous periods will adjust back to equilibrium over the next periods. Similarly the speed of adjustment in investment (δ_4) is expected to be negative as disequilibrium in investment in the previous period should have a positive effect on investment in the next period; that is if the actual flow of housing investment is lower than that predicted by fundamentals, investment will rise.

Both systems of equations, the long-run relationships given by Eqs. (7) and (8) and the short-run relationships given by Eqs. (9) and (10), are jointly estimated using seemingly unrelated regressions (SUR), to account for heteroskedasticity and contemporaneous serial correlations in the error terms across equations.

3.2. Data

The estimation period differs across countries depending on data availability; however, the typical timeframe is from the 1980s to the mid-/late-2000s. Data definitions and sources are as follows. Real prices are inflation-adjusted nominal prices sourced from the OECD or from national data sources. The measure of housing supply is real residential investment, more precisely defined as gross fixed capital formation in the housing sector expressed in volume indices, sourced from the OECD. The stock of housing is the total number of dwellings in a given country sourced from the United Nations Economic Commission for Europe (UNECE). The other variables involved in the estimation include, real income per capita, interest rates, population and construction costs, which are sourced from different OECD databases. Income is measured by either GDP per capita, deflated by the national GDP deflator, or by real household disposable income, depending on data availability. Interest rates are measured by the long or short-term real interest rate (to the extent that households are short-sighted; the short-term interest rate can influence their decision to buy a house), depending on the predominant type of mortgage loan interest readjustment in each country. Population is measured by the share of the 25–44 years old cohort in total population, or the total national population in that age cohort depending on which variable gives the best model fit. A detailed description of the variables and their sources is provided in the Data Appendix.

Fig. 1 plots the evolution of real house prices over the sample period for the OECD countries included in the analysis. Real house prices rose strongly in a majority of OECD countries since the mid-1980s. In several countries, prices have increased by more than 90% since the early 1980s (e.g. Ireland, Spain, the United Kingdom, the Netherlands, Belgium etc.). However, in a few countries real house prices remained stable or even decreased (e.g. Japan, Switzerland and Germany etc.).

However, the increase in real house prices was accompanied by increased housing investment in several countries. Fig. 2 plots the real value of residential investment, which comprises major additions and renovations as well

as new units, for the countries for which data are available. From the mid-1990s until 2006 investment grew rapidly in Spain, Ireland and the Nordic countries, while it was stagnant or even declining in Germany, Switzerland, Japan and Austria. Despite the fact that the data used in the analysis do not allow to distinguish between the different components of residential investment, other sources indicate that in all countries new construction tends to constitute the largest share of housing investment –80% on average in 2004 in countries for which data are available (Bulletin Housing). Only in Sweden, the United Kingdom and Poland does maintenance and repairs of existing dwellings account for at least 30% of investment, possibly reflecting a relatively old stock of housing in these countries.

4. Empirical results

Tables 1 and 2 below report estimation results for the 21 OECD countries, for which the data are available, showing the results for the long-run and short-run price and investment relationships respectively. Panel A summarises the results for the long-run relationships, while Panel B summarises the estimates for short-run dynamics.

As a first step, the order of integration of the series involved in the estimation of the long-run relationship is verified using the standard Augmented Dickey Fuller (ADF) test for the presence of unit roots. The tests include a constant, thus assuming that the overall mean is not zero, and a deterministic trend. The optimal lag length was chosen based on the Modified Akaike Information Criterion (MAIC) of Ng and Perron (2001). The results reported in the Appendix Table A1 provide evidence that, with a few exceptions,³ the long run relationships are integrated of order one. While the use of quarterly data can mitigate the well known low power of such unit root tests in short data samples, some caution is needed in the interpretation of these unit root results. Alternative Dickey-Fuller GLS tests, with higher power and better overall performance in terms of sample size than the standard ADF, yielded similar results. Then, the existence of a long-run relationship between real house prices or investment and the explanatory variables in Eqs. (7) and (8) is verified including the error correction terms derived from the long-run relationship into the dynamic regressions (9) and (10). Each of the error correction terms has a negative sign and is statistically significant, suggesting that Eqs. (7) and (8) can indeed be interpreted as long-run relationships. In the standard Engle Granger approach the existence of a co-integration relationship between the series would be tested by verifying the stationarity of the residuals from the long-run relationships. In this set-up the long-run relationships are, however, estimated employing seemingly unrelated regressions (SUR) rather than OLS, so the standard ADF tests of the residuals are not valid. The alternative test based on the significance of the error correction term in the short-run equation is, therefore, used (see Banarjee et al., 1993).

³ The only exception being that the population and price series tend to be stationary in levels for Italy, Spain and the United Kingdom, and investment is stationary in levels for Australia and United States.

Nominal prices deflated by the consumer priceindex, index, 2000=100

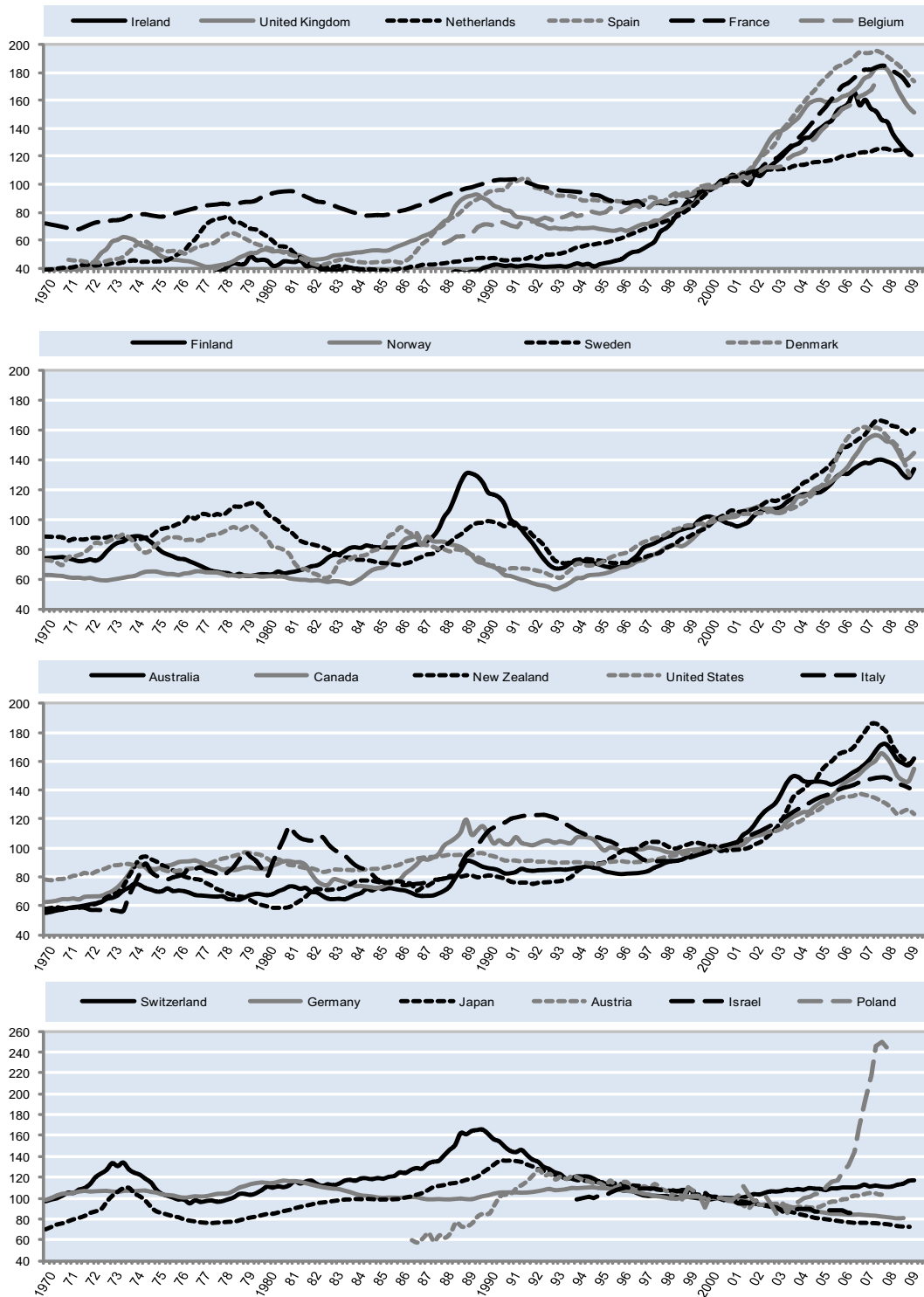


Fig. 1. Real house prices over time. Sources: National sources and OECD Economic Outlook No. 87.

Volume indices, Index, 2000=100, logarithm scale

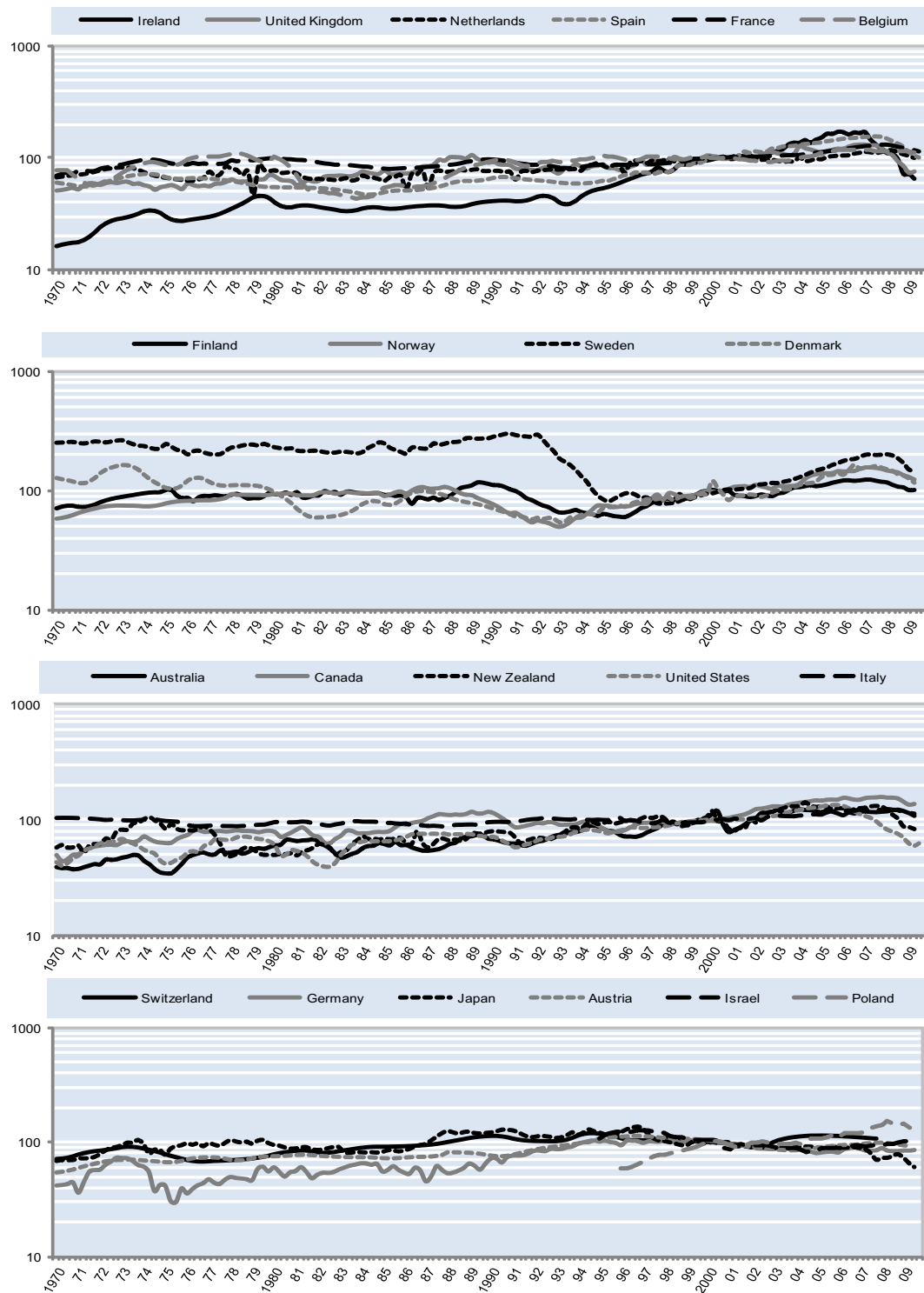


Fig. 2. Residential investment over time. Source: OECD Economic Outlook No. 87.

Table 1Demand equations by country: regression results.^a Source: Authors' estimates.**A. Long run house price relationships by country**

Dependent variable: log real house prices											
	Australia ²	Austria ³	Belgium	Canada	Denmark	Finland	France	Germany	Ireland	Israel	Italy
Income t	1.942 (0.208)***	1.104 (0.386)***	1.613 (0.129)***	1.531 (0.224)***	2.905 (0.629)***	1.634 (0.207)***	4.355 (0.374)***	0.663 (0.152)***	0.615 (0.078)***	0.798 (0.173)***	1.762 (0.511)***
Dwelling stock t	-9.093 (0.738)***	-2.050 (0.309)***	-2.199 (0.329)***	-1.093 (0.249)***	1.811 (0.784)**	-14.928 (2.394)***	-7.332 (0.760)***	-1.367 (0.198)***	-0.739 (0.447)	-0.415 (0.167)**	-2.906 (0.491)***
Interest rate t	-0.000 (0.003)	-0.005 (0.008)	0.001 (0.002)	-0.005 (0.004)	-0.013 (0.005)***	-0.001 (0.003)	-0.012 (0.007)*	0.004 (0.002)**	-0.014 (0.003)***	-0.003 (0.004)	-0.009 (0.005)*
Population t	13.510 (1.114)***	2.669 (1.920)	16.233 (0.836)***	0.700 (0.171)***	..	-1.208 (1.783)	3.488 (1.341)***	1.076 (0.148)***	6.366 (1.422)***	-0.074 (0.121)	..
Constant	-80.875 (4.606)***	-13.534 (6.544)**	-154.667 (5.433)***	-33.107 (4.143)***	18.291 (11.994)	-225.570 (93.819)**	-40.848 (3.315)***	-2.768 (1.526)	-53.291 (8.915)***	0.213 (3.322)	5.301 (10.552)
Number of observations	109	69	93	101	109	69	89	151	109	48	53

B. Short run house price relationships by country

Dependent variable: 1st difference log real house prices											
	Australia ²	Austria ³	Belgium	Canada	Denmark	Finland	France	Germany	Ireland	Israel	Italy
Δ Income t	0.550 (0.240)**	-0.131 (0.896)	0.736 (0.338)**	0.364 (0.208)*	0.074 (0.241)	0.285 (0.250)	0.280 (0.226)	0.232 (0.071)***	1.025 (0.228)***	0.859 (0.493)*	0.261 (0.377)
Δ Dwelling stock t	-8.882 (1.637)***	-1.494 (0.551)***	0.183 (0.642)	0.342 (0.798)	0.204 (0.398)	-2.002 (4.549)	-0.360 (1.370)	0.665 (0.394)*	0.436 (0.416)	0.137 (0.565)	-0.959 (0.859)
Δ Interest rate t	0.002 (0.001)**	0.001 (0.004)	0.000 (0.001)	0.003 (0.001)**	-0.002 (0.001)**	0.000 (0.001)	0.000 (0.001)	0.000 (0.000)	-0.001 (0.000)**	-0.002 (0.003)	-0.000 (0.001)
Δ Population t	9.276 (4.077)**	-0.924 (8.125)	17.278 (4.870)***	-0.166 (0.663)	..	8.180 (2.932)***	0.993 (2.667)	1.036 (0.203)***	2.720 (2.019)	2.875 (5.569)	..
ECT t-1	-0.058 (0.029)**	-0.299 (0.078)***	-0.110 (0.052)**	-0.054 (0.029)*	-0.027 (0.015)*	-0.130 (0.048)***	-0.054 (0.027)*	-0.035 (0.013)***	-0.071 (0.029)**	0.017 (0.031)	-0.050 (0.027)*
Number of observations	108	68	92	100	108	68	88	150	108	42	52
Estimation period	1982q2:2009q2	1986q4:2003q4	1984q4:2007q4	1980q4:2005q4	1980q4:2007q4	1988q4:2005q4	1983q4:2005q4	1970q2:2007q4	1980q4:2007q4	1995q2:2009q1	1990q4:2003q4

(continued on next page)

Table 1 (continued)

A. Long run house price relationships by country										
Dependent variable: log real house prices										
	Japan	Netherlands	New Zealand	Norway	Poland	Spain	Sweden	Switzerland	United Kingdom	United States
Income t	2.015 (0.116)***	2.516 (0.298)***	0.982 (0.498)**	1.191 (0.067)***	3.483 (0.679)***	1.600 (0.148)***	2.825 (0.168)***	3.289 (0.268)***	3.585 (0.387)***	0.832 (0.262)***
Dwelling stock t	-2.875 (0.181)***	-6.606 (0.331)***	-15.186 (2.799)***		-5.955 (2.219)**	1.411 (0.649)**	-3.007 (0.235)***	-3.381 (0.202)***	-11.055 (1.129)***	-0.207 (0.479)
Interest rate t	-0.003 (0.003)	-0.005 (0.002)**	0.002 (0.007)	0.004 (0.003)	-0.002 (0.004)	-0.006 (0.003)**	-0.007 (0.003)***	-0.008 (0.004)*	-0.003 (0.004)	-0.009 (0.004)*
Population t	0.917 (1.332)	14.219 (1.370)***	24.199 (4.221)***		-0.176 (39.285)		4.326 (2.037)**	11.038 (1.260)***	17.205 (2.265)***	
Constant	-9.606 (12.458)	-118.456 (7.330)***	-108.570 (12.877)***	-16.317 (1.168)***	5.609 (441.582)	-37.394 (4.658)***	-49.770 (3.536)***	-28.162 (3.243)***	-133.893 (19.964)***	-7.163 (1.630)***
Number of observations	97	108	54	122	31	130	133	97	101	113
B. Short run house price relationships by country										
Dependent variable: 1st difference log real house prices										
	Japan	Netherlands	New Zealand	Norway	Poland	Spain	Sweden	Switzerland	United Kingdom	United States
Δ Income t	0.105 (0.051)**	0.841 (0.236)***	0.518 (0.205)**	0.347 (0.155)**	2.241 (2.140)	-0.043 (0.228)	-0.126 (0.120)	0.224 (0.341)	1.501 (0.366)***	0.182 (0.113)
Δ Dwelling stock t	0.965 (1.061)	-1.871 (0.801)**	-0.296 (3.476)		2.991 (4.258)		-1.878 (0.803)**	-4.262 (3.488)	4.466 (5.587)	0.715 (0.446)
Δ Interest rate t	0.002 (0.000)***	-0.001 (0.001)	0.001 (0.001)	0.000 (0.000)	-0.001 (0.002)	0.001 (0.000)**	0.000 (0.000)	-0.001 (0.001)	0.000 (0.001)	0.001 (0.001)*
Δ Population t	1.357 (1.257)	30.625 (8.548)***	13.924 (3.797)***		521.143 (328.478)	-0.709 (2.322)	4.722 (3.751)	8.204 (6.369)	25.567 (6.245)***	
ECT t-1	-0.027 (0.013)**	-0.079 (0.035)**	-0.085 (0.027)***	-0.043 (0.014)***	-0.776 (0.218)***	-0.032 (0.011)***	-0.034 (0.014)**	-0.048 (0.029)*	-0.123 (0.024)***	-0.040 (0.014)***
Number of observations	96	107	53	121	30	129	132	96	100	112
Estimation period	1981q4:2005q4	1981q1:2007q4	1994:q3:2007:q4	1979q1:2009:q2	1999q2:2006q4	1977:Q1:2009:q2	1975q4:2008:q4	1980q4:2004:q4	1980q4:2005q4	1980q4:2008q4

^a Results from the long-run and short-run price relationships, estimated jointly with the investment equations, employing a seemingly unrelated regression (SUR) model. All the specifications include a full set of quarterly dummies. Standard errors are reported in parentheses. ***, **, * denote significance at the 1%, 5% and 10%.

^b The stock is lagged 1 year.

^c The model for Austria includes, in addition to the current stock, the stock lagged 1 year because the fit of the model is superior. The coefficients for the 1 year lag are significant at the 1% but are not reported for brevity.

Table 2Supply equations by country: regression results.^a Source: Authors' estimates.**A. Long run residential investment relationships by country**

Dependent variable: log real residential investment											
	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Ireland	Israel	Italy
Real house price t-1	0.528 (0.054)***	0.234 (0.057)***	0.315 (0.028)***	1.187 (0.198)***	1.206 (0.053)***	0.988 (0.037)***	0.363 (0.055)***	0.428 (0.172)**	0.631 (0.045)***	0.379 (0.185)**	0.258 (0.050)***
Real construction cost t-1	-0.678 (0.416)	-2.600 (0.854)***	1.851 (0.274)***	-0.276 (0.272)	-0.410 (0.132)***	-0.313 (0.273)	0.287 (0.218)	1.923 (0.255)***	-0.413 (0.291)	0.743 (0.108)***	-0.275 (0.118)**
Population t	1.697 (0.290)***	6.574 (1.208)***	13.530 (0.747)***	-0.288 (0.319)	3.837 (1.583)**	-0.130 (0.166)	1.366 (0.293)***	5.764 (1.360)***	9.775 (1.819)***	-1.947 (0.121)***	3.459 (1.161)***
Constant	10.612 (1.223)***	-25.001 (7.486)***	9.719 (1.277)***	23.126 (3.177)***	20.019 (0.656)***	20.452 (0.818)***	6.912 (3.601)*	-50.541 (15.836)***	19.512 (1.010)***	39.432 (1.096)***	-13.165 (12.943)
Number of observations	109	69	93	101	109	69	89	151	109	48	53

Panel B: Short run residential investment relationships by country

Dependent variable: 1st difference log real residential investment											
	Australia	Austria	Belgium	Canada	Denmark	Finland	France	Germany	Ireland	Israel	Italy
Δ Real house price t-1	1.205 (0.208)***	0.04 -0.035	0.244 (0.163)	0.392 (0.152)***	0.749 (0.174)***	0.803 (0.132)***	0.441 (0.092)***	0.466 (0.758)	0.238 (0.118)**	0.061 (0.159)	0.102 (0.234)
Δ Real construction cost t-1	-0.628 (0.537)	0.723 (0.356)**	0.690 (0.315)**	-0.083 (0.258)	-0.725 (0.456)	0.457 (0.469)	0.023 (0.120)	1.246 (0.648)*	-0.028 (0.192)	-0.935 (6.014)*	-0.282 (0.260)
Δ Population t	0.291 (1.765)	18.743 (3.815)***	14.520 (3.189)***	1.130 (3.229)	-4.944 (9.202)	-3.736 (2.909)	-8.280 (5.093)	13.697 (10.394)	16.366 (12.617)	-10.961 (0.159)	10.389 (6.609)
ECT t-1	-0.166 (0.048)***	0.068 (0.027)**	-0.105 (0.047)**	-0.058 (0.034)*	-0.216 (0.068)***	-0.310 (0.094)***	-0.052 (0.024)**	-0.200 (0.052)***	-0.113 (0.047)**	-0.323 (0.097)***	-0.622 (0.127)***
Number of observations	108	68	93	100	108	68	88	150	108	42	52
Estimation period	1982q2:2009q2	1986q4:2003q4	1984q4:2007q4	1980q4:2005q4	1980q4:2007q4	1988q4:2005q4	1983q4:2005q4	1970q2:2007q4	1980q4:2007q4	1995q2:2009q1	1990q4:2003q4

(continued on next page)

Table 2 (continued)

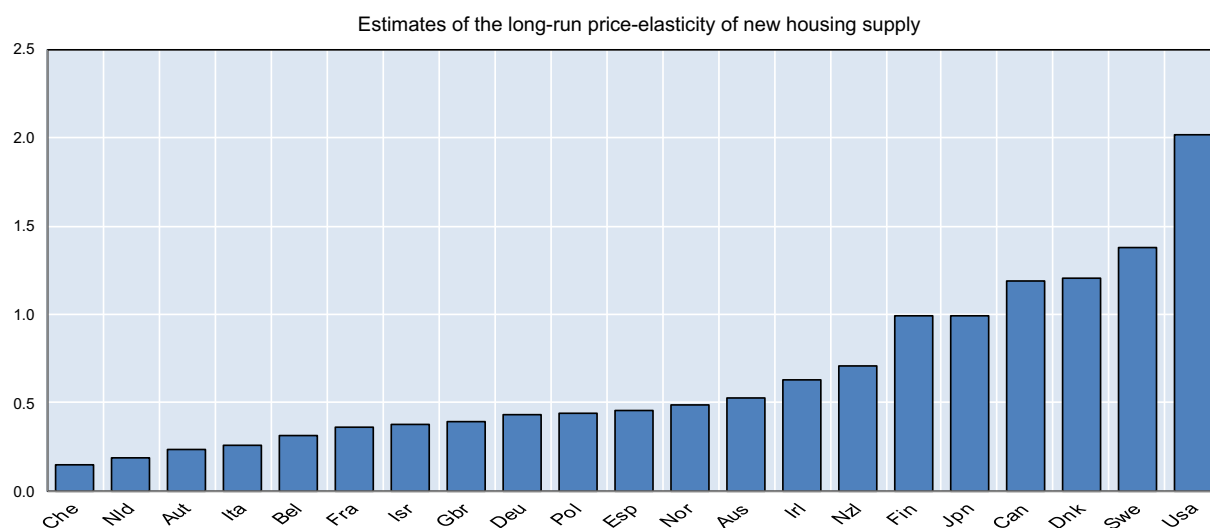
A. Long run residential investment relationships by country

Dependent variable: log real residential investment										
	Japan	Netherlands	New Zealand	Norway	Poland	Spain	Sweden	Switzerland	United Kingdom	United States
Real house price t-1	0.993 (0.074)***	0.186 (0.061)***	0.705 (0.145)***	0.486 (0.024)***	0.442 (0.122)***	0.452 (0.051)***	1.381 (0.193)***	0.146 (0.055)***	0.395 (0.029)***	2.014 (0.376)***
Real construction cost t-1	0.059 (0.230)	-0.542 (0.208)**	-1.040 (0.381)***	-1.039 (0.116)***	-1.275 (0.782)	0.126 (0.192)	-0.275 (0.497)	-0.066 (0.020)***	0.374 (0.078)***	-3.459 (0.600)***
Population t	1.425 (0.391)***	3.016 (0.384)***	0.537 (0.691)	..	-6.566 (6.800)	0.024 (0.371)	36.232 (5.713)***	7.485 (1.356)***	3.935 (0.916)***	6.909 (0.795)***
Constant	9.068 (4.258)**	-3.647 (3.572)	19.527 (5.120)***	27.534 (0.575)***	29.500 (2.701)***	21.103 (3.860)***	10.108 (1.631)***	20.740 (0.380)***	20.029 (0.443)***	-66.341 (10.903)***
Number of observations	97	108	54	122	31	130	133	97	101	113

B. Short run residential investment relationships by country

Dependent variable: 1st difference log real residential investment										
	Japan	Netherlands	New Zealand	Norway	Poland	Spain	Sweden	Switzerland	United Kingdom	United States
Δ Real house price t-1	0.668 (0.342)*	0.470 (0.284)	1.411 (0.530)***	0.399 (0.119)***	0.079 (0.093)	0.236 (0.080)***	0.479 (0.158)***	0.040 (0.041)	0.810 (0.237)***	2.497 (0.396)***
Δ Real construction cost t-1	-0.204 (0.519)	0.482 (0.335)	0.241 (0.721)	-0.162 (0.156)	0.593 (0.546)	-0.053 (0.228)	-0.185 (0.277)	0.003 (0.012)	0.137 (0.098)	-1.625 (0.713)**
Δ Population t	-7.258 (7.946)	26.029 (28.812)	24.244 (17.089)	..	10.251 (14.382)	3.169 (3.387)	5.001 (8.285)	-0.465 (1.624)	6.201 (9.610)	5.038 (11.250)
ECT t-1	-0.136 (0.041)***	-0.775 (0.094)***	-0.429 (0.113)***	-0.162 (0.039)***	-0.300 (0.126)**	-0.076 (0.030)**	-0.126 (0.018)***	-0.087 (0.014)***	-0.277 (0.078)***	-0.072 (0.031)**
Number of observations	96	107	53	121	30	129	132	96	100	112
Estimation period	1981q4:2005q1	1981q1:2007q4	1994q3:2007q4	1979q1:2009q2	1999q2:2006q4	1977q1:2009q2	1975q4:2008q4	1980q4:2004q4	1980q4:2005q4	1980q4:2008q4

^a Results from the long-run and short-run investment relationships, which are estimated jointly with the price equations, employing a seemingly unrelated regression (SUR) model. All the specifications include a full set of quarterly dummies. Standard errors are reported in parentheses. ***, **, * denote significance at the 1%, 5% and 10%.



1. Estimates of the long-run price elasticity of new housing supply where new supply is measured by residential investments (*i.e.* the coefficient on lagged prices in the long-run investment equation as reported in Table 1). All elasticities are significant at least at the 10% level. The estimation period is from early 1980s to mid-2000s. In the case of Spain, restricting the sample to the period 1995–2007, which would reflect recent developments in housing markets (such as the large stock of unsold houses resulting from the construction boom starting in 2000 and peaking in 2007–09), only slightly increases the estimate of the elasticity of housing supply from 0.45 to 0.58.

Fig. 3. Price responsiveness of housing supply: selected countries. Source: Authors' estimates.

4.1. Long-run estimates

Most explanatory variables have coefficients that are significant and consistent with priors. In the price equations (Table 1), income and the variable capturing the structure of the population have a positive effect on prices. House prices tend to increase with households' disposable income and population as income and population growth generates higher demand for housing. As the coefficients in Table 1 indeed show, in the long-run, the elasticity of real house prices with respect to real income are close to or above unity in most countries.

Turning to the effect of the housing stock and real interest rates, they have a dampening effect on house prices. The housing stock tends to have a negative and significant effect on prices, in line with the theoretical prediction that increases in the stock of housing, and thus larger supply, lead to lower prices in the long-run.⁴ The small effect of interest rates on prices is in line with previous evidence suggesting that declining interest rates have had a modest impact on real house prices after controlling for other demand and supply factors (See Andrews et al., 2011). In the supply equation (Table 2), lagged prices and the demographic variable have a positive effect on investment while construction costs have a negative effect. Real construction

costs have, however, a positive and significant effect on investment in Belgium, Germany and the United Kingdom, which could be accounted for poor quality data on construction costs.

Fig. 3 plots the coefficient on lagged prices in the long-run investment equation, *i.e.* the price-elasticity of new housing supply which captures the responsiveness of new supply to changes in prices. There are remarkable differences in the responsiveness of new housing supply across countries. In some countries new housing supply appears to be quite responsive to prices. Supply elasticities in Sweden, the United States, Denmark, Canada, Japan and Finland are at, or above, unity implying that in response to a demand shock housing output will increase proportionally more than prices. On the other hand, there are several countries with particularly low elasticities of supply, including Switzerland, the Netherlands, Austria and Italy. Housing supply is particularly unresponsive to prices in Switzerland relative to other countries, about three times less responsive than in countries like France or Germany, and more than ten times less responsive than in the United States and Sweden where supply is the most responsive among the countries in the sample. These estimates are in line with the scarce cross-country evidence available, which is discussed in detail in the next section.

4.2. Short-run estimates

Turning to the short-run adjustment equations, in the price equation the growth rates of income and population have a positive effect on price inflation, while increases in the housing stock lead to house price deflation. The effect of interest rates changes on price is typically non significant and when significant yields a counterintuitive positive

⁴ The stock of housing is, however, positive and significant in 2/20 specifications (*i.e.* Spain and Denmark). A possible reason for this counterintuitive result is that the stock of housing is picking up the effect of population on house prices, which is not included in the equation as population is highly correlated with the dwelling stock. As a robustness check, including population divided by the dwelling stock, gives a negative and significant coefficient for Spain and non significant for Denmark, while the long-run price elasticities of new housing supply are close to the baseline results.

effect on price growth for some countries (5 of them), which may indicate that the estimation framework is unable to control for the potential simultaneity bias between interest rates and house prices. The coefficients on the error correction term in the price equation are significant and range between -0.027 and -0.776 , suggesting that there are wide differences across countries in the implied speed of price adjustment.⁵ Prices adjust fast to shocks in countries like Poland and Austria, where almost 100% of the gap between actual and equilibrium price is closed within a year, whereas the reaction of prices is slower in, for instance, Japan and Denmark where approximately 10% of the gap is closed within a year.

On the supply side, estimates of the short-run investment equation for the different countries indicate that the growth in lagged prices has a significant positive effect on the growth of investment. However, the other explanatory variables are only significant for few countries. Changes in construction costs and population do not typically influence changes in investment, which may be explained by the fact that typically in the short-run changes in construction costs and population are not large. The coefficients on the error-correction term have the expected negative sign and are significant ranging from -0.052 (France) to -0.775 (the Netherlands).⁶ These estimates imply that between 20% and over 100% of the differences between actual and equilibrium investment are closed within a year depending on the country. There appears to be a positive relationship between the price-responsiveness of housing supply and the speed of adjustment of investment for countries with a very responsive housing supply (e.g. the United States and Sweden) and for countries with a low responsiveness of housing supply (e.g. Switzerland and Belgium). The relationship between the long-term responsiveness of housing supply and the speed of adjustment is more tenuous for the rest of countries. On average investment and prices appear to adjust in similar fashion to shocks.

4.3. Comparison of the findings relative to other studies

Various studies have estimated the price elasticity of housing supply for a reduced number of countries (See e.g. Vermeulen and Rouwendal, 2007a for a review), mostly focusing on the United States and the United Kingdom. Even though, as discussed above, the stock-flow model is the theoretical framework typically used to analyse the housing sector, its empirical implementation widely differs

across studies. This implies that comparisons of estimates across studies are problematic due to differences in analytical frameworks, specification of the supply function (stock, new dwellings or construction permits etc.) and period coverage. Even so, evidence from other studies can serve as a rough benchmark for the estimates of the long-term elasticity of supply reported here. Table 3 summarises these elasticity estimates, together with the speed of adjustment of investment estimates to facilitate the discussion.

In order to draw comparisons between the estimates reported in Table 3 and other studies, countries can be grouped into three categories depending on how price-responsive supply is:

Highly responsive: New housing supply appears to be highly responsive to prices in the United States, Canada, Sweden and Denmark. With the above-mentioned caveats in mind, the available evidence, broadly, confirms our findings. For the United States, the empirical evidence is somewhat mixed with the magnitude of the estimated elasticity depending on the estimation period and analytical framework. Whereas a number of earlier studies find evidence of perfectly elastic supply (Follain, 1979; Malpezzi and MacLennan, 2001; Muth, 1960; Stover, 1986), in general more recent studies tend to find estimates ranging from 0.5 to 3 (e.g. Blackley, 1999; Di Pasquale and Wheaton 1994; Harter-Dreiman, 2004; Poterba, 1984; Mayo and Sheppard, 1996; Topel and Rosen, 1988). It is, however, not straightforward to infer from these studies that there has been a decrease in the elasticity of supply in the United States over time, since the estimation approach differs substantially across studies, potentially affecting the comparability of the findings. For instance, Follain (1979), Muth (1960) and Stover (1986) combine housing supply and demand into a single reduced-form flow equation. While several of the more recent studies estimate the supply of housing relying on a stock-adjustment approach similar to the one in this paper. Comparing both approaches, Malpezzi and MacLennan (2001) find that based on a stock-adjustment model the elasticity estimate for the United States is lower (in the range of 1–6) than that obtained based on a flow model. Given that adjustment in housing markets takes time, for instance due to the durable nature of housing, time to construct and high transaction cost, estimation approaches that explicitly take this into account, such as the stock adjustment model, may be preferable. In any case, from an international comparative perspective, the United States shows a high degree of responsiveness of housing supply to price changes. For Sweden, Hufner and Lundsgaard (2007) also find a high responsiveness of new housing supply (1.4 compared to 1.38 found in this study). By contrast, for Denmark, evidence reported in Swank et al. (2002) points to a much lower elasticity of supply (0.66) than the one found in this study. The difference in the estimates may stem not only from the different estimation frameworks but also from the much longer period covered by our study.

Fairly unresponsive: Housing supply appears to be quite unresponsive to prices in Switzerland and the Netherlands and fairly rigid in Austria, Italy, Belgium, France and the United Kingdom. Evidence for this group of countries is

⁵ The size of the error correction term in the price equation for Poland seems implausibly large (-0.776) implying that 300% of the difference between the actual and the equilibrium price is closed within a year. The error-correction term in the price equation for Israel is not significant, which could be due to the short time series data available.

⁶ Austria stands out as the only country with a positive yet statistically significant adjustment term in the short run investment equation, implying basically no adjustment. A possible reason for this result may be high construction costs, given that the time to obtain a building permit is somewhat longer in Austria than in other OECD countries (Andrews et al., 2011). There is also some evidence that during the nineties there was lack of competition in the Austrian construction industry (OECD, 2001). Although the regression is well behaved; it could also be due to the short time series of available data.

Table 3

Price responsiveness of housing supply and speed of supply response. Source: Authors' estimates.

	Price responsiveness of housing supply ^a	Speed of supply response ^b
Australia	0.528 (0.054)***	−0.166 (0.048)***
Austria	0.234 (0.057)***	0.068 (0.027)**
Belgium	0.315 (0.028)***	−0.105 (0.047)**
Canada	1.187 (0.198)***	−0.058 (0.034)*
Denmark	1.206 (0.053)***	−0.216 (0.068)***
Finland	0.988 (0.037)***	−0.310 (0.094)***
France	0.363 (0.055)***	−0.052 (0.024)**
Germany	0.428 (0.172)**	−0.200 (0.052)***
Ireland	0.631 (0.045)***	−0.113 (0.047)**
Israel	0.379 (0.185)**	−0.323 (0.097)***
Italy	0.258 (0.050)***	−0.622 (0.127)***
Japan	0.993 (0.074)***	−0.136 (0.041)***
Netherlands	0.186 (0.061)***	−0.775 (0.094)***
New Zealand	0.705 (0.145)***	−0.429 (0.113)***
Norway	0.486 (0.024)***	−0.162 (0.039)***
Poland	0.442 (0.122)***	−0.300 (0.126)**
Spain	0.452 (0.051)***	−0.076 (0.030)**
Sweden	1.381 (0.193)***	−0.126 (0.018)***
Switzerland	0.146 (0.055)***	−0.087 (0.014)***
United Kingdom	0.395 (0.029)***	−0.277 (0.078)***
United States	2.014 (0.376)***	−0.072 (0.031)**

^a Coefficient on lagged prices in the long-run investment equation.^b Coefficient on the error correction term in the short-run investment equation.

only available for the United Kingdom and the Netherlands. It suggests that housing supply is particularly inelastic in these countries, translating into poor supply responses to demand shocks and strong rises in house prices (e.g. André, 2010; Barker, 2004; Vermeulen and Rouwendal, 2007a,b). While econometric evidence on the price responsiveness of housing supply for Austria, Italy, Belgium and France is lacking, Figs. 1 and 2 show that prices have risen more than investment in these countries over the recent past, suggesting relatively low supply responsiveness.

Responsive: A last group of countries display intermediate housing supply responsiveness, including New Zealand, Ireland, Australia, Norway and Spain. Evidence for this group of countries is only available for New Zealand and Spain. For New Zealand, Grimes (2007) finds a elasticity of housing supply between 0.5 and 1.1, in line with the evidence reported here (0.71). For Spain, Taltavull de la Paz (2008) finds that, despite the great amount of residential construction carried out since the mid-1990s, the long-run elasticity of housing supply over a longer time period (from 1987 to 2004) is not as high as could be expected – around 0.47 – and slightly lower than the one found here of 0.57 over 1977–2009.

Most studies on the responsiveness of housing supply are conducted at the national, and in some limited cases regional levels, as opposed to local housing markets. Using more disaggregated data is desirable, as it can help to uncover, for instance, variations in the timing of housing cycles across local markets or inter-metropolitan area movements in population. Nonetheless, data constraints typically difficult such analyses. Some evidence, however, exists on differences in supply elasticity across local housing markets in the US. For instance, Saiz (2010) finds based on US metropolitan data that elasticity ranges from 0.60 in

Miami to 5.45 in Wichita, and that differences in supply elasticity are a function of both land-use regulations and natural geography. This confirms the finding by Gyurko and Saiz (2006) that land markets explain the wide divergence in housing supply elasticity, and ultimately, home values across US metropolitan areas. Some evidence also exists for Australia, where Mc Laughlin (2012) finds that there is a significant variation in housing supply elasticity, both across housing types, as well as across capital cities.

5. Which factors influence the responsiveness of housing supply to prices?

Cross-country differences in the long-run price responsiveness of housing supply are related to both policy and non-policy factors. First, geographical and demographic conditions, such as physical limitations on land for development, can restrict the supply of land in certain areas and have a negative impact on the supply of housing in the long run. Indeed, a simple cross-country correlation shows that the estimated housing supply elasticity is lower in more densely populated countries (Fig. 4, Panel A). The same appears to be true within countries. For instance, based on existing estimates by Green et al. (2005) on the responsiveness of housing supply for US cities, Panel B in the same figure suggests that supply is more rigid in cities with a greater population density.

But, government policies can also influence housing supply. For instance, land-use and planning policies are intended to reduce negative externalities that can be associated with new housing construction, but if they are poorly designed they may also restrict supply responsiveness. For instance, Fig. 5 shows that housing supply responsiveness tends to be lower in countries where it takes longer to

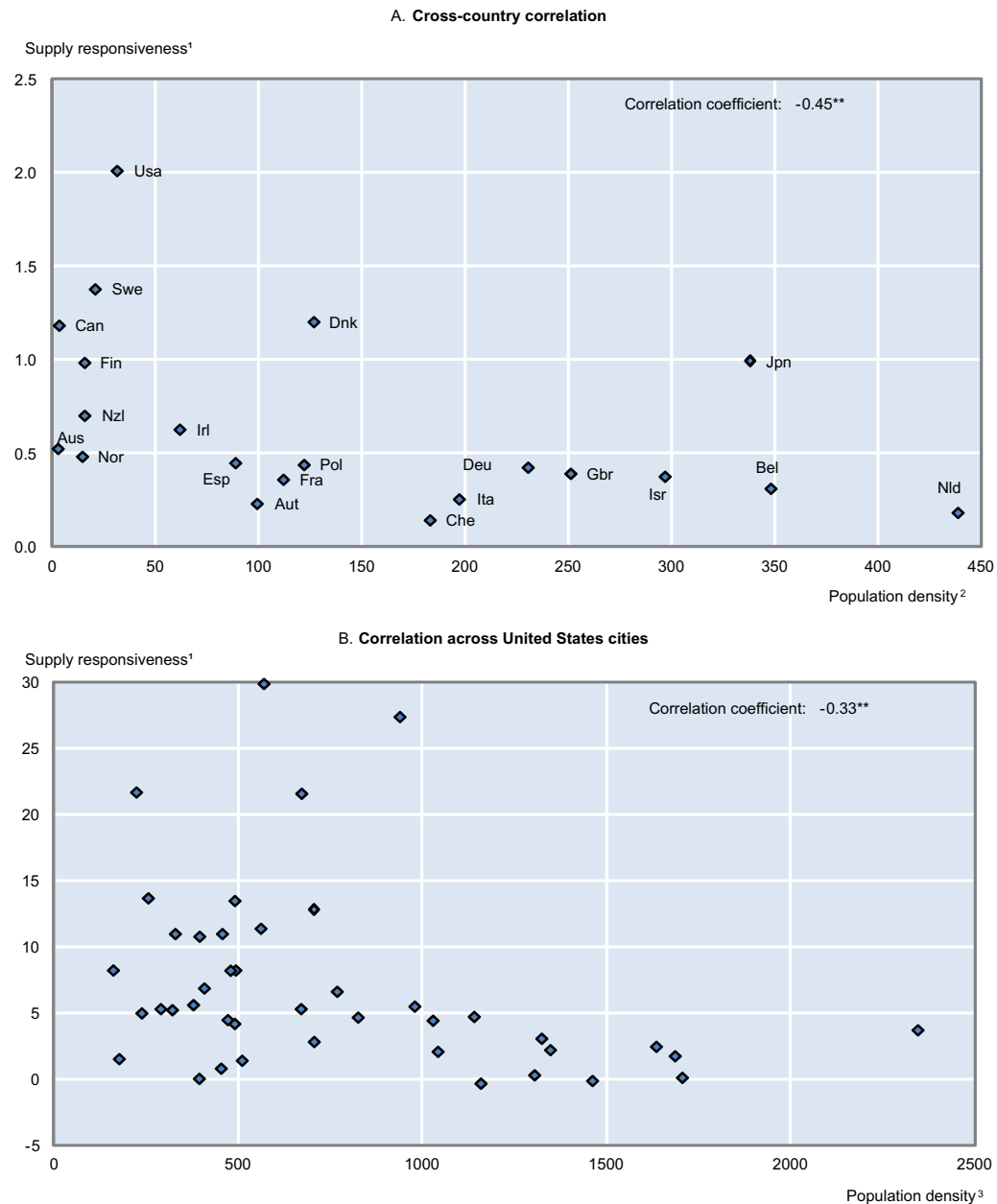


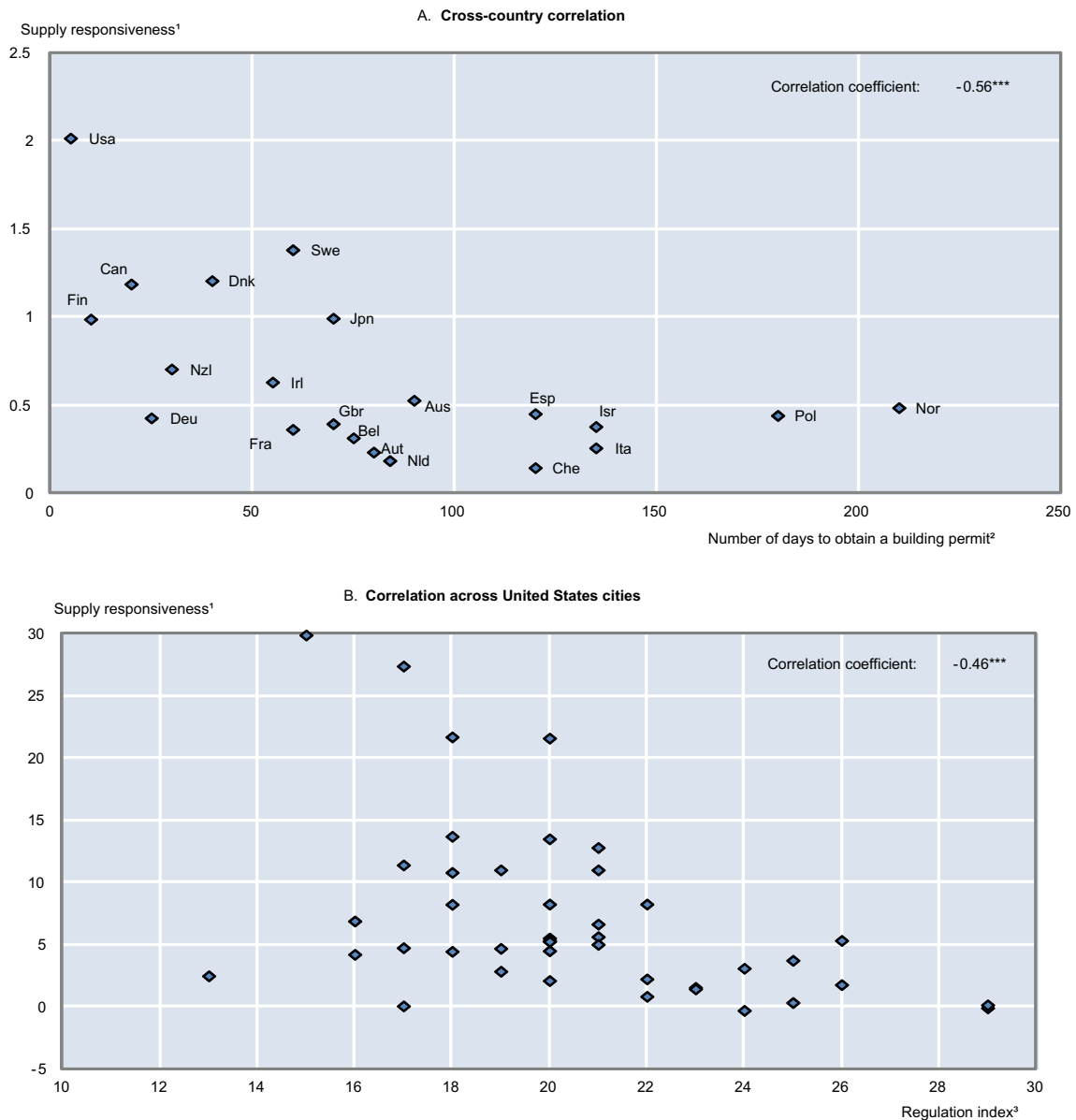
Fig. 4. Price responsiveness of supply and scarcity of land. Sources: Authors' estimates, United Nations Population Database and Green *et al.* (2005).

acquire a building permit.⁷ Interpreting this correlation causally, an efficient design and enforcement of land-use

⁷ This correlation is robust to controlling for scarcity of land in a simple regression explaining the elasticity of supply with land-use regulation and scarcity of land. In addition, the correlation is also evident if the lifespan of building permits are used as a regulatory measure instead of the waiting time for obtaining a building permit.

regulation would have the potential for making housing supply more responsive to prices. This illustrative finding is also evident across cities in the United States, providing further support for the notion that housing supply responsiveness is related to regulations on land-use and planning.

However, while there is an emerging consensus that local land-use regulations in some countries have become a binding constraint on the supply of new housing units,



1. Authors' estimates of country-specific supply responsiveness and estimates of supply responsiveness for United States cities taken from Green *et al.* (2005). *** Denotes significance at the 1% level.

2. The number of days to obtain a building permit is obtained from the World Bank Doing Business indicators.

3. The land-use regulation index captures approval time of building permits, available land for residential housing and access to adequate infrastructure and ranges between 7–35 (increasing in regulation) (see Malpezzi, 1996).

Fig. 5. Price responsiveness of supply and land-use regulations. Sources: Authors' estimates, World Bank Doing Business indicators, Malpezzi (1996) and Green *et al.* (2005).

there is much less of a consensus on the magnitude of the impact (e.g., Gyourko, 2009 for an overview). Apart from regulations on land-use, the provision of infrastructure and other public services complementary to housing, such as road junctions or water drainage, is also likely to influence supply, though hard evidence of this link is not available (e.g. Barker, 2008, for a discussion).

Housing supply responsiveness is also potentially affected by the degree of competition in the residential con-

struction industry (Barker, 2004). Available studies typically find that average mark-ups in the construction industry are low relative to other non-manufacturing industries (Molnár and Bottini, 2010; Bouis and Klein, 2009), although the extent of competition varies across countries (See Andrews *et al.*, 2011). In most countries, the construction industry is typically characterised by a large number of relatively small firms. However, thinking of construction as an atomistic industry may be misleading

as it is often the case that only a limited number of contractors are capable of managing large construction projects. In such cases competition may be low (OECD, 2008). In light of this evidence, implementing an effective competition policy that enforces anti-trust regulation and hinders collusive behaviour in the construction sector seems to be important for improving the responsiveness of housing supply.

Finally, a number of studies illustrate the adverse effects of poorly-designed rent regulations on housing supply (e.g. Arnott, 1995; Ellingsen and Englund, 2003). Stringent rent regulations potentially discourage new construction and maintenance by capping the price of rentals, thus lowering the net return on residential investments for rental purposes and constraining housing supply (Arnott, 2003; Sims, 2007). In line with this, an illustrative correlation analysis in Andrews et al. (2011) shows that across countries, stricter rent controls tend to be associated with lower quantity and quality of rental housing, as measured by the share of tenants lacking space and those reporting sub-standard housing in terms of a leaking roof.

6. Conclusion

This paper estimates the elasticity of new housing supply for 21 OECD countries shedding light on the differences in supply conditions. Results suggest that supply responsiveness is strong in North America and some Nordic countries, while it is comparatively low in Switzerland, the Netherlands and Austria. A responsive housing supply is important to avoid bottlenecks in different segments of the market in response to an increase in housing demand. However, the flip side is that in flexible-supply countries, housing investment adjusts more rapidly to large changes in demand. This contributes to more cyclical swings in economic growth. Despite this trade-off, in the longer term a more flexible supply of housing is desirable as it allows a better match of housing construction to changes in housing demand patterns across the territory.

Housing supply can be made more responsive through policy reforms, notably in the areas of housing regulations and taxation, which can help to avoid excessive increases in house prices and their negative impact on the economy. Efficient land-use policies and regulations can ensure a better use of land in countries where scarcity of land restricts supply and pushes prices up. In addition, in some countries there may be scope for streamlining cumbersome licensing processes to facilitate a flexible adjustment of housing supply to demand conditions.

From a taxation perspective, well-designed taxes on under-used or vacant land can be imposed on landowners to encourage residential development in countries with a shortage of land for residential construction. For example, linking the assessment of property value for tax purposes to the market value may increase incentives for developing vacant land (Johansson et al., 2008). In countries where the construction industry is characterised by a few large constructors, the enforcement of competition policy which hinders collusive behaviour is also important for a flexible supply. The design of such policies should, however, bal-

ance the benefit of additional supply against the potential cost of new developments in terms of congestion and environmental amenity losses.

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Appendix A. Data Appendix

A.1. House prices

House prices sourced from the OECD house price database and national data sources (for Israel) deflated by the consumer price index, sourced from the OECD Analytical Database (ADB).

A.2. Construction costs

Construction costs for residential investment sourced from the OECD Main Economic Indicators (MEI) database and national data sources (for Australia, Japan, New Zealand and Israel) deflated by the GDP deflator, sourced from the OECD ADB.

A.3. Dwelling stock

A measure of the housing stock was computed for each country based on information on the total number of dwellings in each country sourced from the United Nations Economic Commission for Europe (UNECE). Data for Australia and Japan was not available from UNECE and was therefore sourced from the Reserve Bank of Australia and the Japan Statistics Bureau, respectively.

A.4. Short term interest rates

Real short-term interest rate based on the private consumption deflator sourced from the OECD ADB database.

A.5. Long term interest rates

Real long-term interest rate based on the private consumption deflator sourced from the OECD ADB database.

A.6. Residential investment

Gross fixed investment in housing sourced from the OECD ADB database.

Table A1Augmented Dickey Fuller test.^a Source: Authors' estimates.

	Australia		Austria		Belgium	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-0.869	-4.792***	-2.943	-1.682*	-0.729	-1.447*
Log residential investment	-5.301***	-6.746***	-2.010	-4.490***	-3.048	-3.068***
Log GDP	-2.487	-4.274***	-3.175*	-6.021***	-3.187	-3.548***
Log GDP per capita	-2.054	-4.161***	-2.551	-5.922***	-2.325	-3.433***
Log disposable income	-1.148	-4.358***	-2.320	-2.974***	-3.236	-3.332***
Log disposable income per capita	-0.722	-4.148***	-2.204	-2.839***	-3.203*	-3.404***
Weighted average short term and long term interest rates	-1.700	-4.508***	-1.982	-4.334***	-2.720	-9.074***
Short term interest rates	-1.700	-4.508***	-2.677	-8.112***	-2.058	-3.215***
Long term interest rates	-1.726	-4.070***	-2.159	-4.387***	-2.764	-8.780***
Log real construction costs	-2.516	-4.299***	-2.798	-6.349***	-2.383	-7.375***
Log total population	-1.422	-1.888*	-3.596**	-0.600	-1.420	-1.194
Log population aged 25-44	-3.680**	-1.087	-4.119***	-1.192	-3.616***	-0.460
Share of the total population aged 25-44	-3.962**	-0.562	-2.042	-0.634	-4.059***	-0.541

	Canada		Denmark		Finland	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-2.776	-2.822***	-1.724	-3.730***	-2.572	-3.985***
Log residential investment	-2.219	-6.125***	-1.887	-4.273***	-2.993	-3.509***
Log GDP	-2.238	-3.432***	-2.436	-3.278***	2.665	-3.912***
Log GDP per capita	-2.262	-3.374***	-1.972	-3.231***	-2.638	-3.887***
Log disposable income	-3.592**	-3.783***	-2.248	-4.579***	-2.114	-3.911***
Log disposable income per capita	-3.410*	-3.832***	-3.136	-4.484***	-2.124	-3.864***
Weighted average short term and long term interest rates	-2.072	-8.374***	-2.910	-4.855***	-2.202	-9.314***
Short term interest rates	-1.577	-5.854***	-2.371	-3.660***	-2.2	-9.295***
Long term interest rates	-2.079	-8.249***	-1.897	-6.224***	-2.313	-9.263***
Log real construction costs	-2.590	-4.130***	-3.008	-5.028***	-1.425	-2.652***
Log total population	-0.956	-1.652*	-4.567***	-2.523***	-1.429*	-1.202
Log population aged 25-44	-3.615**	-1.405	-0.817	-0.010	-3.513***	-1.370*
Share of the total population aged 25-44	-4.041***	-0.889	-1.741	-0.476	-1.370	-1.387*

	France		Germany		Ireland	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-3.090	-2.752***	-1.644	-3.415***	-1.767	-3.491***
Log residential investment	-2.524	-3.722***	-2.252	-5.522***	-1.804	-4.173***
Log GDP	-2.830	-4.208***	-1.546	-4.153***	-1.833	-2.211***
Log GDP per capita	-2.544	-4.277***	-1.698	-4.303***	-1.864	-1.336*
Log disposable income	-3.086	-4.371***	-0.652	-2.851***	-1.361	-3.755
Log disposable income per capita	-3.118	-4.363***	-0.450	-3.007***	-1.408	-1.922**
Weighted average short term and long term interest rates	-1.724	-7.461***	-2.591	-5.720***	-1.625	-6.021***
Short term interest rates	-1.588	-7.400***	-2.230	-5.884***	-1.924	-5.958***
Long term interest rates	-1.816	-7.411***	-2.591	-5.720***	-2.217	-4.354***
Log real construction costs	-0.869	-2.339***	-2.676	-3.526***	-2.544	-8.761***
Log total population	-2.113	-2.789***	-4.154***	-1.356	-6.646***	-1.302
Log population aged 25-44	-3.491***	-0.249	-3.135***	-0.524	-4.807***	-0.440
Share of the total population aged 25-44	-3.799***	-0.329	-3.285***	-0.595	-3.519***	-0.555

	Israel		Italy		Japan	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-3.535*	-2.592**	-3.848**	-2.957***	0.223	-4.391***
Log residential investment	-0.294	-3.206***	-2.867	-4.146***	-0.686	-2.966***
Log GDP	-2.077	-2.839***	-0.767	-2.207***	0.070	-2.823***
Log GDP per capita	..	..	0.181	-2.121***	-0.093	-3.511***
Log disposable income	..	..	-1.830	-2.628***	-1.225	-2.987***
Log disposable income per capita	..	..	-1.310	-2.647***	-1.397	-3.084***
Weighted average short term and long term interest rates	..	..	-1.825	-6.029***	-1.709	-4.627***
Short term interest rates	-2.925	-3.416***	-1.813	-6.027***	-1.685	-4.695***
Long term interest rates	..	..	-2.182	-5.566***	-1.806	-4.717***
Log real construction costs	-1.181	-1.945**	-2.613	-4.103***	-2.086	-1.986**
Log total population	-0.582	-2.596**	-3.237***	-1.519	-1.671	-6.205***
Log population aged 25-44	..	..	-3.185***	-0.612	-3.292*	-2.866***
Share of the total population aged 25-44	..	..	-3.952***	-0.876	-4.967***	-1.580

(continued on next page)

Table A1 (continued)

	Netherlands		New Zealand		Norway	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-2.508	-2.920***	-3.078	-3.242***	-1.977	-3.049***
Log residential investment	-2.300	-8.000***	-2.633	-3.141***	-2.058	-3.569***
Log GDP	-1.751	-3.061***	-2.584	-2.738***	-1.513	-2.921***
Log GDP per capita	-1.846	-9.183***	-2.570	-2.835***	-0.830	-2.695***
Log disposable income	-2.796	-3.003***	-3.637	-2.040**	-1.380	-5.136***
Log disposable income per capita	-2.704	-3.059***	-3.361*	-2.035**	-1.613	-5.158***
Weighted average short term and long term interest rates	-1.729	-7.962***	-3.082	-8.842***	-3.590**	-4.801***
Short term interest rates	-2.002	-6.583***	-2.222	-6.828***	-3.590**	-4.801***
Long term interest rates	-1.780	-7.949***	-3.082	-8.842***	-1.452	-7.142***
Log real construction costs	-2.395	-4.512***	-1.646	-6.348***	-2.227	-4.860***
Log total population	-0.277	-2.192**	-6.100***	-1.923	-1.827	-1.867**
Log population aged 25-44	-3.773***	-0.036	-4.061***	-0.081	-3.806**	0.256
Share of the total population aged 25-44	-5.789***	-0.599	-5.735***	-1.443	-3.192*	0.370

	Poland		Spain		Sweden	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-0.546	-2.533***	-3.162*	-2.919***	-1.302	-2.596***
Log residential investment	-3.095	-1.689*	-2.810	-1.673**	-2.908	-2.973***
Log GDP	-3.776**	-6.463***	-2.678	-3.101***	-2.322	-3.772***
Log GDP per capita	-4.24***	-6.382***	-2.305	-2.954***	-2.614	-4.821***
Log disposable income	-4.490***	-10.041***	-2.748	-2.986***	-1.249	-3.675***
Log disposable income per capita	-4.442***	-10.088***	-2.700	-2.894***	-1.371	-3.755***
Weighted average short term and long term interest rates	-1.357	-4.514***	-5.151***	-8.785***	-5.635***	-10.446***
Short term interest rates	-1.357	-4.514***	-2.517	-5.996***	-8.506***	-17.735***
Long term interest rates	-2.223	-1.358*	-2.784	-11.571***	-1.172	-5.826***
Log real construction costs	-2.790	-2.362**	-1.981	-7.195***	-2.672	-3.008**
Log total population	-6.151***	-1.384*	-6.101***	-2.010**	-3.832**	-1.796**
Log population aged 25-44	-3.320*	-1.595**	-2.254	-1.787**	-3.690**	-2.312**
Share of the total population aged 25-44	-4.297***	-1.978**	-2.505	-1.304*	-3.021	-1.368*

	Switzerland		United Kingdom		United States	
	Level	1st difference	Level	1st difference	Level	1st difference
Log real house prices	-2.795	-3.624***	-3.601**	-3.449***	-1.088	-3.396***
Log residential investment	-1.925	-4.212***	-2.621	-3.655***	-4.297***	-4.007***
Log GDP	-3.673**	-3.969***	-2.977	-3.323***	-2.392	-4.166***
Log GDP per capita	-3.721**	-4.014***	-3.047	-3.268***	-2.096	-4.292***
Log disposable income	-3.367*	-3.203***	-2.440	-4.514***	-2.207	-4.572***
Log disposable income per capita	-3.249*	-3.278***	-2.215	-4.482***	-1.892	-4.679***
Weighted average short term and long term interest rates	-3.135	-7.471***	-1.866	-6.108***	-2.524	-7.798***
Short term interest rates	-3.135	-7.471***	-1.844	-6.103***	-2.803	-9.152***
Long term interest rates	-4.225***	-3.567***	-3.030	-9.047***	-2.530	-7.483***
Log real construction costs	-2.004	-3.905***	-1.901	-4.437***	-0.544	-4.532***
Log total population	-5.226***	-1.275	-0.684	-0.648	-6.637***	-1.751**
Log population aged 25-44	-1.444	-0.429	-1.410	-0.084	-5.061***	-1.378*
Share of the total population aged 25-44	-1.345	0.256	-1.403	0.158	-4.492***	-1.599*

^a ***Stationary at 1% level of significance. **Stationary at 5% level of significance. *Stationary at 10% level of significance.

A.7. Population 25–44

Population aged between 25–44 years old sourced from the United Nations Population database.

A.8. Total population

Total national population sourced from the United Nations Population database.

A.9. GDP

Gross Domestic Product in volume, at 2005 PPP US dollars, sourced from the OECD ADB database.

A.10. Disposable income

Net household disposable income in value sourced from the OECD ADB database.

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